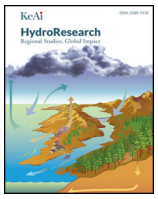
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Addressing water scarcity challenges through rainwater harvesting: A comprehensive analysis of potential zones and model performance in arid and semi-arid regions–A case study on Purulia, India

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ABSTRACT

Water scarcity in arid and semi-arid regions is a critical global concern, necessitating innovative solutions to address increasing water demands in these vulnerable areas. This study focuses on tackling this challenge by identifying and classifying rainwater harvesting zones based on their potentiality and comparing the performance of two machine learning models, Artificial Neural Network (ANN) and Random Forest (RF), for optimizing rainwater harvesting strategies. The study area is Purulia, a district in India. Extensive literature review was conducted to identify key factors influencing rainwater harvesting. Open-source remotely sensed data were employed to pinpoint rainwater harvesting potential zones. A multi-criteria decision-making technique was applied to assess the importance of various factors. Results indicated that rainfall, slope, runoff potential, soil, land cover, and drainage density are the six crucial factors for selecting suitable rainwater harvesting locations. Approximately 2% of the area is unsuitable, 8% is poorly suitable, 33% is moderately suitable, 45% is highly suitable, and the remaining 12% is extremely suitable in Purulia. Two predictive models were developed, with the RF algorithm demonstrating nearly 99% accuracy. Finally, remedial techniques for mitigating water scarcity through rainwater harvesting are discussed separately for urban and rural areas. This research article embraces a comprehensive approach to address water-related concerns, offering a replicable framework applicable globally, with a specific focus on arid and semi-arid regions.

1. Introduction

Arid and semi-arid regions reside within high pressure belt zone where chance of rainfall is minimum (Katiraie-Boroujerdy et al., 2017). Rainfall deficit causing drought is one of the important characteristics of arid and semi-arid regions (ASAR). Rainfall deficiency affects the natural resources and livelihood of that region. Water scarcity creates a negative impact on economic development also (Morante-Carballo et al., 2022). The impact of global climate change has further exacerbated the water utilisation challenges in ASARs (Yifru et al., 2021). Several studies have been depicted that, within limited scope of water resources optimum utilisation of water is the ultimate option and channelisation of rainfall generated water in superlative way can be the ultimate solution for this region that can lead to sustainability (Boers and Ben-Asher, 1982; Bruins et al., 1986; Helmreich and Horn, 2009; Kahinda et al., 2007; Wei et al., 2005).

According to researchers, rainwater harvesting can be the best possible way to create a resilience against this massive problem of water scarcity specifically for ASAR (Ammar et al., 2016). Identification of best suitable sites for rain water harvesting is one of the intriguing topics for researchers worldwide (Adham et al., 2016; Hatibu et al., 2006; Mahmoud et al., 2016). Identification of best possible zone for rain water harvesting depends on various common physical factors and other socio-economic factors and the importance of the variables differs from place to place for ASAR. From the beginning of regional planning different methods had been applied to identify Rainwater harvesting potential zone (RWHPZ). Recently researchers are taking help of remote sensing techniques and geographic information system (GIS) to incorporate every possible variable that can lead to identify RWHP zones with minimum data and maintaining cost effectiveness (Debebe et al., 2022; Mahmoud and Tang, 2015; Waghaye et al., 2023). As importance of various factors are responsible to create the RWHPZ for any ASAR, Multi criteria decision making (MCDM) approach is one the of the best method that has been applied by several researchers in their study (Al-Ghobari and Dewidar, 2021; Rajasekhar et al., 2020a). Different approaches have already been applied related to MCDM technique (Arunbose et al., 2021; Goswami and Ghosal, 2022; Machiwal

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et al., 2011). Nowadays, identification of potential zones is not limited to the application of MDCM techniques. Further advanced machine learning techniques, and even deep learning techniques being implemented rigorously (Bui et al., 2023; Mohsin et al., 2022; Omeka et al., 2023; Vojtek et al., 2021).

In light of this above observation, it has been identified that, though the RWHPZ has been identified for ASAR but these studies created a site-specific analysis following a complex MCDM technique that lacks in identification of a similar location apart form study area. So, this paper approached for creation of an efficient model to identify RWHPZs that can easily identify the potentiality of rainwater harvesting for any area that fulfils the properties of ASAR. Specifically, the aim of the study is to identify different RWHPZ for an ASAR and create a competent model applying principal variables as input according to its potentiality in respect with rainwater harvesting that can easily identify the rainwater harvesting potentiality for any location with minimum effort. Further resilience related to water scarcity and possible way with maximising utilisation of RWHP zones has been tried to identify exclusively for the location. Certain objectives have been considered to accomplish this study. The primary focus of the study was on determining the most suitable technique, emphasizing the use of GIS methods in conjunction with widely applied MCDM techniques; analytic hierarchy process (AHP) to identify RWHPZs. Then selective machine learning algorithms like artificial neural network (ANN) and random forest (RF) have been applied to create predictive models to identify which is the most accurate one. Furthermore, efficient techniques have been identified separately for urban and rural areas for practising best remedial measures for rain water harvesting.

For this study, Purulia has been selected as study area which is treated as ASAR in India. Here different rudimentary methods related to RWH has been applied but precise categorisation of lands in terms of potentiality and efficient use of the highly potential zones is yet to achieve. In this paper, above mentioned approaches have been followed for the regional development and sustainability. Novelty of this paper lies in the approach for creating an advanced model with efficient machine learning algorithm. This paper also finds site specific solutions related to rain water harvesting. Overall, a holistic approach has been applied in this paper that highlighted every possible way to create a resilience against water scarcity.

2. Study area

Purulia is a district is in the eastern part of India (Fig.1). It is situated in the extreme part of the state West Bengal. Locational existence is limited within 22.79°N to 23.69°N and 85.81°E to 86.90°E. This district is under Chotanagpur plateau. Purulia and its adjacent district considered as 'dryland areas of West Bengal' (Chatterjee, 2015). According to state agricultural plan of West Bengal, this district is considered as arid region (Bhunia et al., 2020). Soil is mainly acidic here and mostly unsuitable for agriculture. Water scarcity is the most important concern here. As the rivers are non-perennial and seasonal, water deficiency reaches up to maximum level at summer season after ground water depletion.

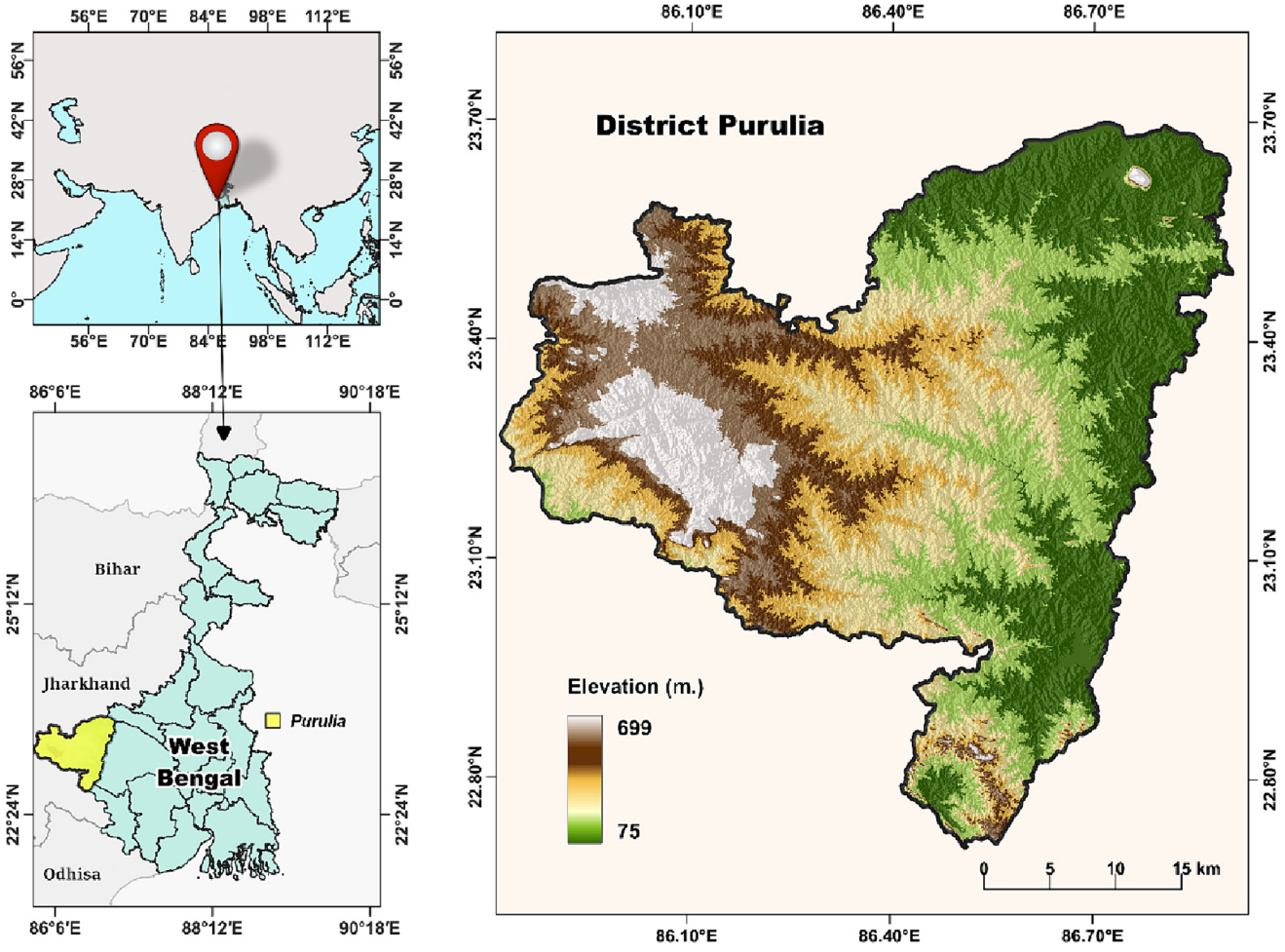


Fig. 1. Location map of the study area.

(Ranganathan et al., 2018). Optimum utilisation of rainwater can solve the severe water crisis in this district.

3. Materials and methods

The current investigation unfolds in two distinct phases. Initially, six parameters were selected through a comprehensive review of literature in the first phase. Subsequently, MCDM techniques were employed to construct an identification map for RWHPZ. Moving to the second phase, machine learning algorithms were employed to develop predictive models (Fig. 2).

3.1. Parameter selection method

As per the objectives of this paper, potentiality of rainwater harvesting has been identified for the study area. The potentiality of rain water harvesting depends on certain criteria. After extensive literature review it has been identified that, some common physical factors are extremely important to identify RWHPZ. Some socio-economic criteria are play an important part for delineation but those are very site specific and mostly used for exclusion purpose. Based on literatures LULC, slope, soil texture, drainage density, runoff coefficient and rainfall have been recognized as the most used criteria for RWHPZ identification. It has been observed that the importance of rainfall as a criterion has been neglected in certain studies. While, in this study rainfall has been considered as most important criteria for RWHPZ identification because it is the most crucial factor in ASAR. Therefore, six criteria have been selected for RWHPZ identification: rainfall, slope, runoff potential, soil texture, landcover, and drainage density (Table 1). The importance of each character has been described in Table 2.

3.2. Data source

For this study, distinctive determination has been given to identify six parameters that lead to identify RWHPZ. Data collection for every parameter needed individual effort. Data sources for above mentioned six parameters have been mentioned below along with other several factors related to the study or preprocessing (Table 3).

3.3. Details about the parameters

3.3.1. Rainfall

Purulia is counted as a discrete area having aridity in nature because the amount of yearly rainfall is comparatively lower than other adjacent areas. Based on 30 years daily rainfall data, it has been observed that, maximum yearly rainfall is observed more than 1500 mm. whereas, in case low yearly rainfall it is lower than 1300 mm. Rainfall is not equal throughout the district. From the Fig. 3a, it can be observed that, rainfall pours maximum in the eastern side and it declines towards the west. On the basis of average yearly rainfall, the study area can be divided into five zone according to availability of rainfall.

3.3.2. Runoff coefficient

Calculation of runoff coefficient is the combined effect of rainfall, land cover, slope and hydrological properties of soil. Researchers mostly followed SCS-CN approach (Table 4) to understand the runoff potential of the study area (Kumar and Jhariya, 2017; Rajasekhar et al., 2020b). In the study, runoff potential map has been created to runoff coefficient for each of the location. It has been identified that; runoff potential is minimum near the hilly area (Fig. 3b). It increases with declination in elevation and runoff potentiality is maximum in areas with gentle slope.

3.3.3. Slope

Geomorphology and topography of Purulia has been reflected on the slope profile (Fig. 4a). Mostly the land is undulating with moderate slope as it is under plateau region. Land with significant angle of slope can be observed in the Ayodhya Hill which is in the south western part. Few discrete areas with extreme slope can be observed on the extreme northern eastern and south eastern part also. Overall slope ranges within 10°.

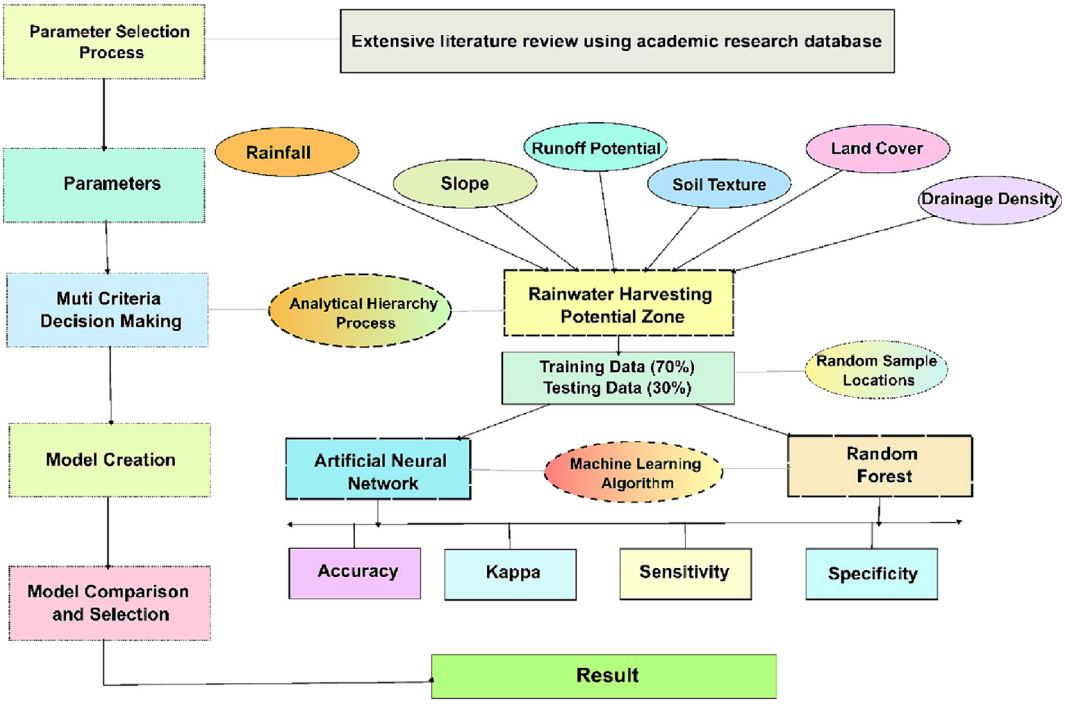


Fig. 2. Methodological framework of the study.

Table 1
Criteria that have been incorporated by authors in their study for identification of RWHPZ.

Authors	LULC	Rainfall	Slope	Drainage density	Soil texture	Runoff coefficient
Roy et al. (2022)	✓		✓	✓	✓	✓
Wu et al. (2018)	✓		✓		✓	✓
Kumar and Jhariya (2017)	✓		✓	✓	✓	
Hashim and Sayl (2021)	✓		✓		✓	✓
Mahmoud and Alazba (2015)	✓	✓	✓		✓	✓
Mahmood et al. (2020)	✓		✓	✓		✓
Singhai et al. (2019)	✓		✓		✓	
Ibrahim et al. (2019)	✓	✓	✓	✓	✓	
Haile and Suryabhagavan (2019)	✓		✓	✓	✓	✓
Rajasekhar et al. (2020b)	✓		✓	✓	✓	
Mandal et al. (2023)	✓		✓	✓	✓	
Kar et al. (2020)	✓		✓	✓	✓	
Shadmehri Toosi et al. (2020)	✓	✓	✓	✓	✓	

3.3.4. Soil texture

Soil texture of this study area is mainly composed of silt and loamy elements. Most part of the study area is covered with clay loam. Significant amount of mixture with sand clay and loam can be found in clustered pattern mostly in the northern part of the district and loamy soil disconnectedly covered the rest of the area (Fig. 4b and 5a).

3.3.5. Land cover

This district mainly underdeveloped district among other districts because of extreme uncomfortable climate. Land cover has been formed accordingly. Mainly dense vegetation and range land are observed around the hilly areas. Seasonal flooded vegetation is observed around rainy season discretely throughout the district near to the availability of water. Most of the land has been covered into agricultural land. Dispersed settlement patterns are observed in built-up area. Bare land is also observed haphazardly according to the landforms here. Availability of waterbodies are minimum and mostly are man-made except few small rivers (Fig. 5b).

3.3.6. Drainage density

Collecting the data from stream order (Fig. 6a), drainage density has been calculated. It has been observed that, Drainage density is maximum with more than 1 km/sq.km in the central part facing west to east. In rest part of this district drainage density of that district is comparatively lower in values (Fig. 6b).

It is noteworthy to mention that, above mentioned six parameters have been further utilised for identification RWHPZ. Mapping of six parameters has been accomplished in QGIS (version 3.28) and creation map has been done based on inverse distance weighted interpolation method wherever it is necessary. For proper visualization of the thematic map of individual parameter, distribution of data has been represented through natural breaks Jenks classification method.

3.4. Multi criteria decision making analysis

Potentiality of RWH for an area is dependent on major principal factors. Each of the factors shares a significant role for optimum utilisation of rain-fed water. Identification of the role of every criterion and calculate its weightage have been calculated using muti criteria decision MCDM making technique. For this Analytical hierarchical process has been followed. As it is the most common and most appropriate method that can lead to generate individual weightage for each of the factor. Already application of analytic hierarchy process (AHP) in case of finding weightage for RWHP is common. A pairwise comparison matrix has been formed between number of criteria that has been applied in this study where relative importance between two criteria has been numbered within scale ranges between 1 and 9 (Table 5). It is noteworthy that, for appropriate consideration of weightage for every parameter, authors visited the study area several times for a clear view about the situation. Further, verification of their understanding they studied extensive number of literatures related to the study area specific to this research domain. After this process whole matrix has been normalised and aggregating the relative importance weight for each criterion has been calculated. Consistency of this analysis needs validation, that has been analysed by consistency ratio. All these processes have been completed in python language platform using *'pydecision'* package.

After calculation of weightage and validation with consistency ratio, overlay analysis has been processed. For this, every variable used in this analysis has been divided into five sub criteria where 1 signifies unsuitability and 5 represents most suitable for rainwater harvesting (Roy et al., 2022). As suitability reflects potentiality of an area, it has been represented as suitability. Further, weighted sum overlay analysis has been processed in ArcGIS 10.8. Malczewski (2006) undertook a process where he took the thematic layers and multiplied them by their respec-

Table 2
Six important criteria for identification of RWHPZ and their reasons behind selection.

Criteria	Importance in RWHPZ identification
Rainfall	Higher values of rainfall add scope for more water collection
Slope	Depending on gentler slope it would be easier to use hold the storm water
Runoff potential	Runoff potential creates a significant role in water holding capacity of an area
Soil texture	Depending on soil texture it is easier to calculate retention capacity
Land cover	Land cover signifies the scope and need for RWH in a specific location
Drainage density	It directly supports the transporting capacity of RWH

Table 3
Sources and links of the datasets used in this study.

Type	Source	Link
Rainfall	Indian meteorological department	https://www.imdpune.gov.in/cmpg/Griddata/Rainfall
Slope	Shuttle Radar Topography Mission (SRTM)	https://www2.jpl.nasa.gov/srtm/
HSG	Ross et al. (2018)	https://daac.ornl.gov/SOILS/guides/Global_Hydrologic_Soil_Group.html
Soil texture	Hengel, 2018	https://opengeohub.org/about-openlandmap/
Landcover	Esri, Sentinel-2 land cover explorer	https://livingatlas.arcgis.com/landcoverexplorer/

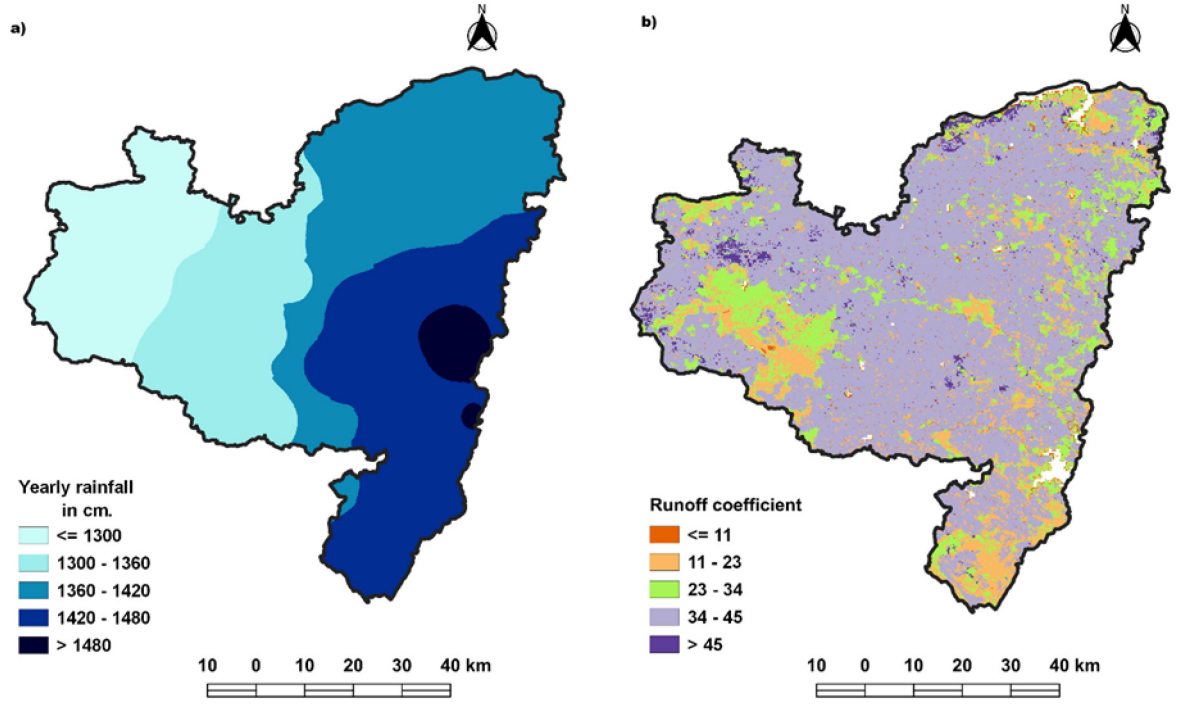


Fig. 3. a) Yearly rainfall map, b) Runoff coefficient map of Purulia.

tive weights. The outcomes were then combined using the subsequent formula (Eq. 1):

$$A_j = \sum_{i=1}^{m} W_i X_{ij} \quad (1)$$

In this context, A_j represents the eventual suitability score within each cell. The symbol X_{ij} corresponds to the suitability level of the ith cell concerning the jth layer, while W_i signifies the standardised weight of the ith layer. After rescaling (0 to 1) the resultant potential zone map was then rearranged into five distinct categories: (a) not suitable (0–0.20), (b) low suitability (0.20–0.40), (c) moderate suitability (0.40–0.60), (d) high suitability (0.60–0.80), and (e) very high suitability (0.80–1).

3.5. Creation of model for suitable rainwater harvesting potential zone

Any machine learning based model needs significant number of samples. Usually, these sample datasets are divided into training and testing purpose. In this paper, primarily random 1467 samples points have been collected from the study area (Fig. 7a). From the collected samples 70% (1025) locations have been selected for training the model and rest of the samples (442) have been used to test the models. a) rainfall, b) slope, c) runoff coefficient, d) soil texture, e) drainage density, and f) landcover has been selected as input dataset and RWHPZ classes has been selected as output dataset of the model. After collecting the samples whole dataset have been standardised within a range of 0 to 1 to eliminate numerical biasness in the model (Fig.7b). As the paper, classifies the RWHPZ through classification model, standardised RWHPZ data have been reclassified into five zones suitability according to potentiality.

Table 4
Values for different curve numbers of different land cover types.

Description	CN_A	CN_B	CN_C	CN_D
Water	1	1	1	1
Trees	57	73	82	86
Flooded vegetation	1	80	87	93
crops	76	85	90	93
built-up	77	86	91	94
Bare ground	77	86	91	94
Rangeland	63	77	85	88

Two individual model has been prepared using supervised machine learning method. It is observed that, labelled data can be accurately classified using this method. This method itself has versatility in terms of its usage (Günen, 2022). Other advantages like, fast convergence, efficient handling of noise and support for incremental learning created appropriate scope to use this model in this study. Classification methods has been used here based on terms of suitability of land to be identified as RWHPZ. Researchers have applied different algorithms like, Linear regression, logistic regression, Decision Trees, SVM, Random Forest, Neural Networks, Gradient Boosting algorithms etc. in the field of spatial analysis (Darabi et al., 2021; Günen, 2022, 2024; Palermo et al., 2020). For the efficiency of this model, ANN and Random Forest both algorithms have been used.

3.5.1. Artificial neural network

Function of human brain has been reflected behind the creation of ANN algorithm (Cherkassky et al., 2006). It is constructed with nodes based upon input data and output data. ANN usually remains organised with artificial neurons with layers and these neurons collectively collects input data and transforms into meaningful output data. In these process certain hidden layers are involved (Chen et al., 2017). This type of algorithm is mainly used for pattern recognition, regression and classification. (Kumar Arora and Mathur, 2001) provided details about the internal process involved in ANN method where net input is transformed by common activation function i.e., sigmoid function and weightage of input connection is determined by backpropagation algorithm. The magnitude of weight impacts the inclination of the activation function. This implies that weights govern the speed at which the activation function initiates, while bias is employed to defer the initiation of the activation function (Ray et al., 2020). In this paper, ANN model has been created with six input variables mentioned above and reclassified five classes of RWHPZ has been provided as an output, three intermediate hidden layers have been created in this model

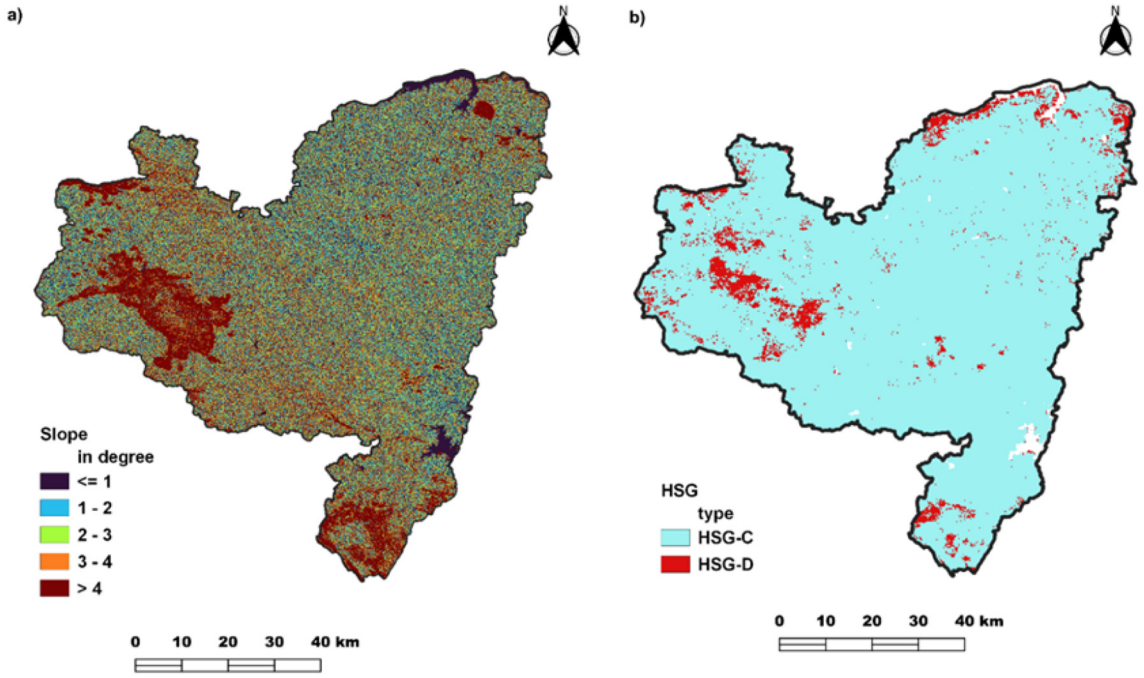


Fig. 4. a) Slope map and b) HSG of Purulia.

where two bias values have been generated (Fig. 8). Resampling process 10-fold cross validation has been repeated for 5 times. Three hidden layers were optimised with decay value of 0.0001. This model has been created with R programming language and for this 'caret' package has been utilised.

3.5.2. Random Forest

Random Forest is a combination of several machine learning algorithm mainly combined with different tree classifiers (Jaiswal and Samikannu, 2017). Breiman (2001) described the random forest process where large number classifier tree is generated those votes for

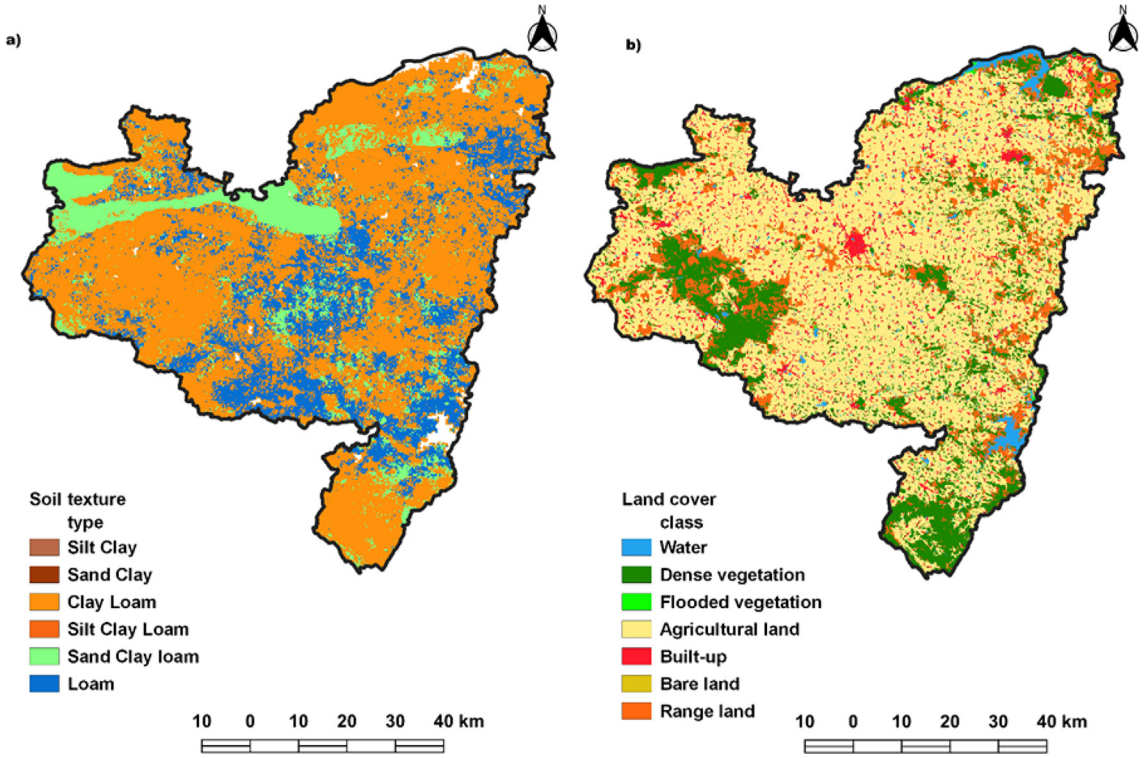


Fig. 5. a) Soil texture map and b) Land cover map of Purulia.

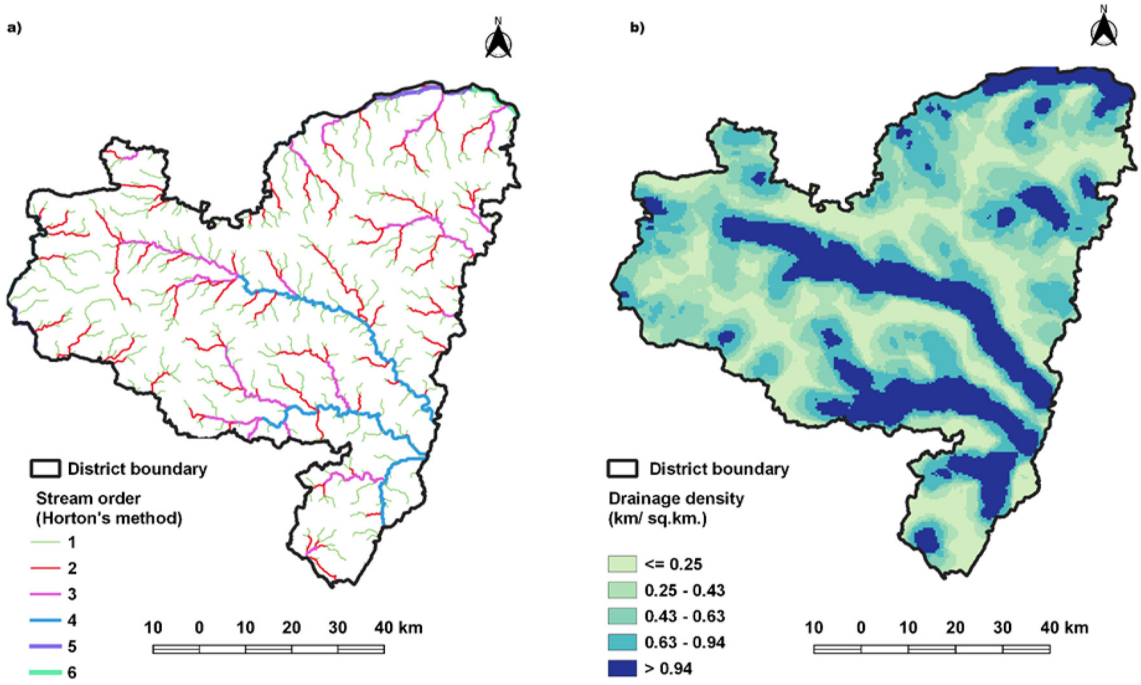


Fig. 6. a) stream order map and b) drainage density map of Purulia.

the most popular class. It is an extension of classification and regression trees (CART) and widely accepted supervised machine learning technique that is used for classification and regression purpose (Ziegler and König, 2014). This method is being significantly used in spatial analysis. It is not only used in remote sensing but for geographic data it efficiently manages large number of diverse input dataset and predict result with optimizing accuracy (Belgiu and Drăguţ, 2016). Effectiveness of this machine learning algorithm depends on tuning number of trees (ntree) and number of variables (mtry) (Naghibi et al., 2017). In this paper, 500 random trees have been used for classification and at each split two variables have been utilised. This model has been programmed with R programming language and for this 'RandomForest' package has been used.

3.6. Validation process

Efficiency of any model reflects through its validation process. In this paper both of the classification model has been evaluated on the basis of same parameters. Understanding the proficiency of any model through confusion matrix is the most common practise that has been applied individually for the models (Marom et al., 2010). Also, overall accuracy and kappa coefficient value has been calculated and compared the results. Further two classification models have been analysed based on its sensitivity (true positive rate) and specificity (true negative rate) values for each class. Further details have been provided in the table for better understanding about the comparison metrices used in the study to compare the efficiency of the models (Table 6).

Table 5
Pairwise matrix along with weightage of six criteria involved in the study.

	Rainfall	Slope	Runoff	Soil texture	Land cover	Drainage density	Weightage
Rainfall	1	2	2	2	3	3	0.29
Slope	0.50	1	2	2	2	3	0.22
Runoff	0.50	0.50	1	2	2	3	0.17
Soil texture	0.50	0.50	0.50	1	3	3	0.15
Land cover	0.33	0.50	0.50	0.33	1	3	0.10
Drainage density	0.33	0.33	0.33	0.33	0.33	1	0.06

4. Result

4.1. Analysis of factors behind identification of potential rainwater harvesting potential zone

Purulia reflects the characteristics of an area near to tropic of cancer. Here geomorphology is undulating in few places. Collecting details about the variables responsible for creating potentiality in rainwater harvesting, it is observed that rainfall is not affluent all over the study area (Fig. 9). Minimum rainfall is observed in the western part annually less than 1300 mm. near about 17% of area are under this region. Rainfall tends to increase from west to east with only 50 mm. change yearly. Near about15% of total area gets rainfall more than 1350 mm, whereas towards east one fifth of area gets less than 1400 mm. rainfall. One third of area in the middle portion of the study area gets rainfall in between 1400 mm to 1500 mm yearly. In the extreme eastern portion only 12% of area having rainfall near about 1500 mm, in a year. Near about 50% of the land is with extremely gentle slope where average slope is less than 2.5° and near about 40% of total area is under 6.5°. Undulation is minimum for near about 90% of the total area. Only 6% of area where land is undulating up to 12.5°. Rest of the areas mostly in western portion are part of plateau region where minor hills are dominating in geomorphology and there slope angle ranges within 22.5° to up to 60°.

Range of runoff potential for the study area is limited within negligible to 56%. Area wise runoff potentiality varies over the study area. Runoff coefficient is maximum in gentle plain agricultural land and minimum in hilly and rangelands. Near about 17% of area where minimum runoff potentiality is observed. Runoff within 28% is observed in 27% of total area. Rainfall runoff percentage within 28 to 38 is observed within minimum of area. High runoff potentiality up to 56% is observed in 13% of the total area.

Characteristics of plateau region is reflected in the soil texture of the study area. Near about 65% of the land is covered with clay and loam and 21% of land is purely covered with loamy properties. Mixture of soil

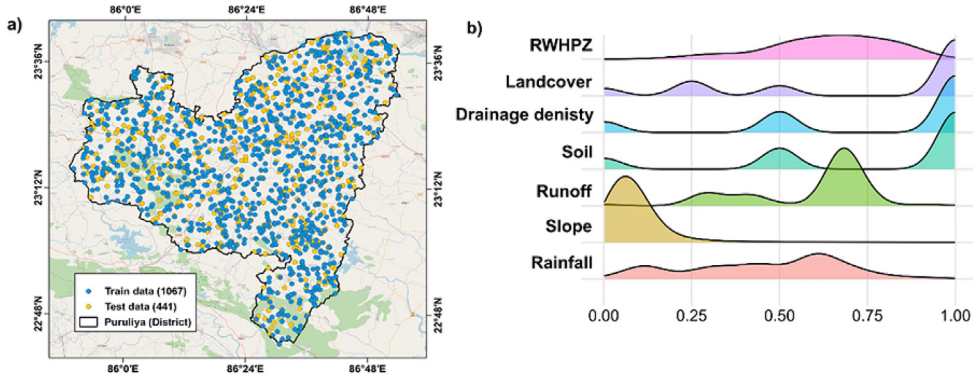


Fig. 7. a) Location of sample points used in the study, b) distribution of rescaled utilised for the model.

sand and loam covered only 12% of area. Here loamy properties of dominates almost total study area.

Structure of drainage network has been reflected in drainage density. It is minimum (0 to 0.33 km/km^2) in 30% of the total area whereas drainage density is maximum (more than 2 km/km^2) in only 5% of the total area. Drainage density carries a range between 0.33 and 2.01 km/km^2 in rest of the 63% of the total area.

Purulia is mainly depends on primary activities and secondary activities. Landcover in this study area dominant over the land use. Near about 56% land are covered by agricultural activities and from them significant amount land is treated as barren land for most of the months in a year. Near about 16% land is covered with trees and vegetation. Specially the hills are surrounded with dense forest. Due to its topography and climatic condition, extensive area of land is observed with native herbaceous vegetation; this land covered almost one fifth of the total area.

4.2. Analysis of rainwater harvesting potential zone

It is observed that, minimum area with less than 2% of the area is unsuitable for rainwater harvesting and mostly these locations are observed in the western portion of the study area (Fig. 10). Only 8% area is counted as poorly suitable where potential of rain water harvesting is near to impossible. One third of area in western part and southern part of Purulia can be counted as moderately potential for rain water harvesting. It is observed that near about 45% of the area is under high suitability and the potential for rainwater harvesting further increase and suitability reach up to highest in the central eastern part of the study area covering less than 12% of the total area.

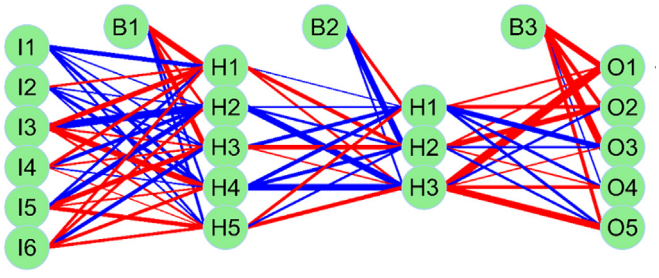


Fig. 8. Network of ANN model where I1 to I6 represents input variables,H1 to H5 and H1 to H3 represents hidden layers, B1, B2 and B3 represent biases and O1 to O5 represents five classes of RWHPZ as output.

4.3. Comparison between ANN model and RF model

Both of the model was created based on same training dataset. After the model creation with train dataset, at first confusion matrix have been compared for the selection of one model (Table 7). Comparing prediction and references it is observed that, RF model optimised the model with better accuracy than ANN model in every five classes. RF model is absolutely accurate in case of identification of unsuitable, poorly suitable, and most suitable classes but in case of ANN model it is not accurate for any of the classes. After creating both the model, accuracy and Kappa value have been compared for training dataset and testing dataset individually (Fig. 11). Comparing the outcome, it has been observed that, in case of ANN model, training dataset showing 88% accuracy whereas with test dataset applying the model, accuracy turned to 83%. The situation is far better in time of application of RF model. Accuracy is near to 99% with training dataset and 89% in time of application of this model with test dataset. After creating confusion matrix, kappa value has been obtained for both model that showing. Kappa values are far better in RF model compared to ANN model. For ANN based model, identification of unsuitable area is poor in comparison with RF model in terms of sensitivity and specificity values (Fig. 12). Result for rest of the classes is comparatively better in case of ANN model but in RF model unsuitable class has been identified most accurately and values of rest of the class for sensitivity and specificity are close to 1. Comparing true positive rate and true negative rate it can easily state that efficiency of RF model is better than ANN model.

5. Discussion

Purulia region is considered as ASAR in India. This district is known for water scarcity specially in summer (April to June) season. Due to poor source of water agricultural prospects diminishes in most time of the year. Economically, it is treated as underdeveloped region. With

Table 6
Details about comparison metrics for validation and comparison of the models.

Comparison metrices	Equation	Remarks
Sensitivity	$\frac{True\ positivies}{True\ positives+False\ negatives}$	proportion of actual positives that are correctly identified by the test.
Specificity	$\frac{True\ negatives}{True\ negatives+False\ positives}$	proportion of actual negatives that are correctly identified by the test.
Accuracy	$\frac{True\ positives+True\ negatives}{Total\ population}$	ratio of correctly predicted instances to the total instances.
Kappa coefficient	$\frac{P_o-P_e}{1-P_e}$	P_o is the observed agreement P_e is expected agreement under chance

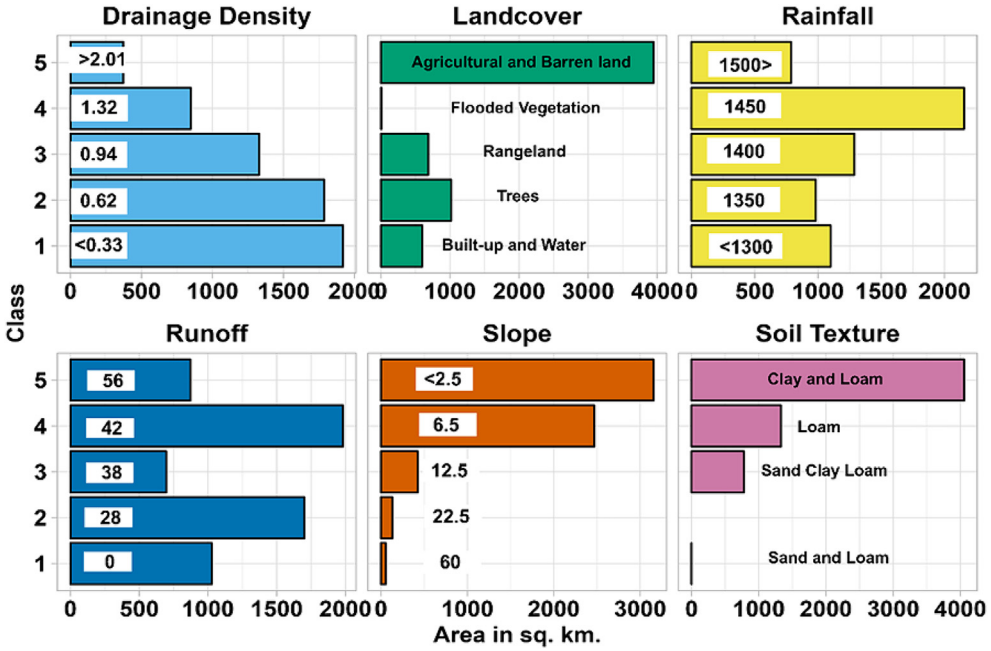


Fig. 9. Criteria wise area for each of the class.

scarcity in groundwater resources, proper usage of rainwater and utilisation of this water for domestic usage and agricultural purpose is beneficial for this region. Rainwater harvesting is the ultimate choice for this region. In this purpose, application of remote sensing and GIS technology helped to gather information for solving this issue. Factors responsible for rainwater harvesting curated using extensive literature review. It is observed that, identified six important factors are carrying a certain importance individually. From the produced map of RWHPZ is observed that, rainfall, drainage density and runoff potential are the crucial factor for identification of RWHPZ. Also, variation of slope soil and land cover played a major role. Eastern part of the region has been identified as most suitable place for rain water harvesting. Already established reservoir on that zone supports the identification process. It has been identified on suitable zones already major and minor dams are employed for optimal utilisation of flood water. Further, two models have been produced individually to identify most efficient model with

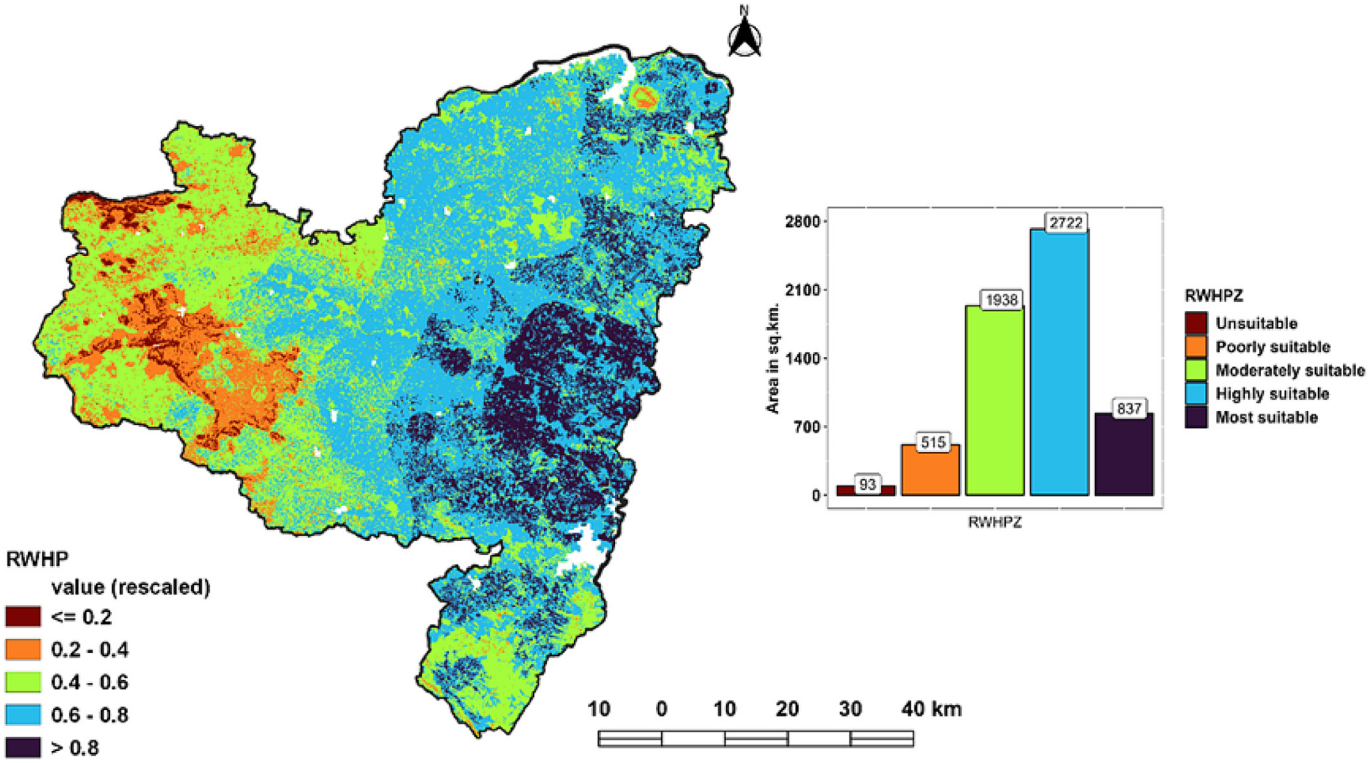


Fig. 10. Rainwater harvesting potential zone map and details about the area.

Table 7
Confusion Matrix for of ANN and RF model.

Prediction	Reference				
	Unsuitable	Poorly suitable	Moderately suitable	Highly suitable	Most suitable
ANN					
Unsuitable	12	2	0	0	0
Poorly suitable	3	81	12	0	0
Moderately suitable	0	7	256	31	0
Highly suitable	0	0	27	410	19
Most suitable	0	0	0	22	143
RF					
Unsuitable	14	0	0	0	0
Poorly suitable	0	95	0	0	0
Moderately suitable	0	1	292	0	0
Highly suitable	0	0	2	456	2
Most suitable	0	0	0	0	163

two most used algorithms in machine learning technique. Both supervised model, showed their efficiency in their own terms. After three stages of validation through confusion matrix, accuracy, specificity, and sensitivity it has been proved that RF model is the most efficient model for Purulia. In Purulia district, after demarcation of most suitable areas some techniques must be followed separately for urban areas and rural areas (Table 8).

In urban areas, rainwater harvesting should be done utilising minimum space. For this in large size buildings recharge pit may be constructed with 1 to 2 m wide and 2 to 3 m deep and mesh should be provided above it to prevent contamination of polluting substances. Recharge trench can be also created with 0.5 to 1 m wide along the side of the residential areas. Recharge wells can be created in official and administrative buildings where runoff potential is maximum. In Purulia urban area covers a minimum portion of land and settlements are in clustered form. For rain water harvesting recharge pit in residential houses and recharge trench around the locality can be optimum solutions in Purulia.

In rural areas, surface spreading techniques are common. With locally available stones gully plug can be constructed. Construction of contour bunds in lands with moderate slope also proves to be efficient for long term sustainability. In high runoff potential area, creation of percolation tank can be effective of irrigation purpose in Purulia. In locations near small streams with gentle slope check dams also can be constructed. Groundwater dams can also be constructed where valley is wide and outlet of the watershed is narrow. This district is mostly

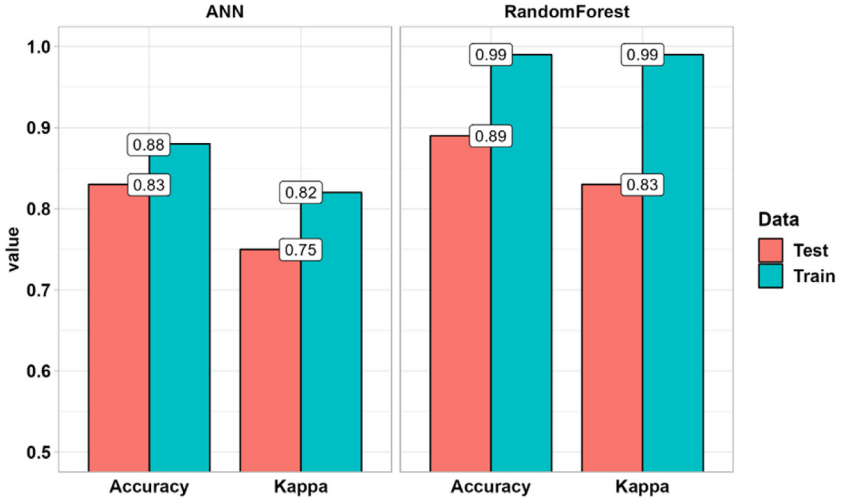


Fig. 11. Accuracy and Kappa value of ANN and RF model with train and test data.

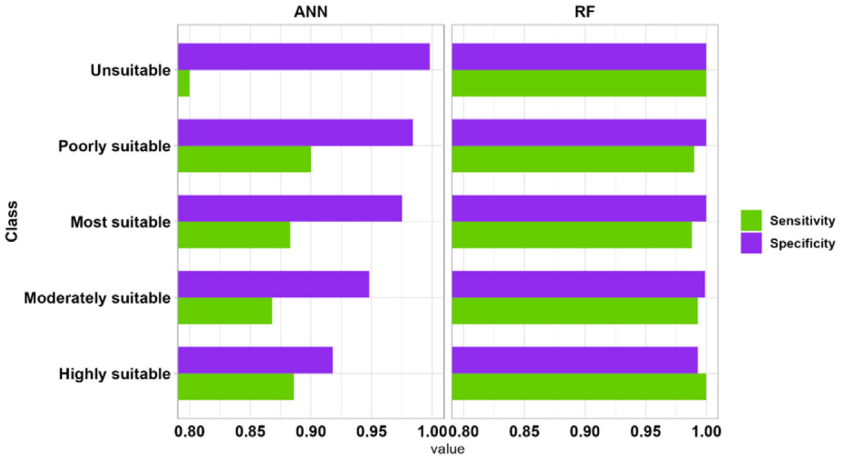


Fig. 12. Sensitivity and specificity values for ANN and RF model for individual classes.

Table 8
Rainwater harvesting techniques for urban and rural areas.

Urban Areas	Rural Areas
Recharge Pit	Gully Plug
Recharge Trench	Contour Bund
Tube well	Percolation Tank
Recharge Well	Check Dam and Groundwater Dams

dependent on agriculture and development of irrigation is the most needful concern here. Already major minor dams along with reservoirs have been constructed but site-specific utilisation of rainwater resource should be applicated. Creating gully plug, contour bunds, percolation tank near the higher drainage density area problems of water scarcity can be solved.

This machine learning model generated with RF algorithm can be appropriately applied to overcome the situation of water scarcity in Purulia. Importance of this model lies in the applicability of it other ASAR over the world. With easily available open-source dataset over the web can be utilised to apply in the model. With minimum effort, a map can be visualised to locate the RWHPZ using primary six criteria. After identification of RWHPZ, different remedial measures need to applied to over the water scarcity and severity of draught in ASAR.

6. Conclusion

This article reflects a way out to solve the issues of water scarcity and optimum utilisation of rainwater specifically in ASAR. Overall study, examined a large area district in arid region, identified factors that can create a better scope for rain water harvesting, classified the district according its potentiality of rain water harvesting and further a model has been produced comparing its efficiency with another model, finally remedial measures have been discussed to apply these techniques over the district. In water scarce areas, practise of rainwater harvesting can make changes in overall quality of life and thus in economy. Applying this model in Purulia, policy makers and planners can locate potential areas for rainwater harvesting and applying remedial measures overall development and growth can be idealised. It is be mentioned that, identification, modelling and remedial measures have been described based on the results. Scope remains open for further critical assessment. Authors of this paper, provided a holistic approach to battle against unfavourable water scarcity situation in ASAR. This process of identification, modelling and remedial measures can be recapitulated in any other ASAR in the world with certain level of expertise in the knowledge domain.

CRediT authorship contribution statement

Subhra Halder: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Suddhasil Bose:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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